Muscular Dystrophy Pedigree Chart
Answer key

Worth 35 points total

Background Information

Pedigree charts are very important to many different fields of science. One reason they are important is because, they help scientists understand the genetic patterns of diseases. It is important to be able to interpret pedigree charts in order to learn the pattern of a disease or condition. Specifically, using a pedigree chart, you can tell if the disease or condition is autosomal, X-linked, dominant, or recessive.

Before you start this activity it is important to review several symbols:

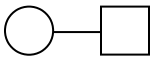
-Unaffected male - Affected male

-Unaffected female -Affected female

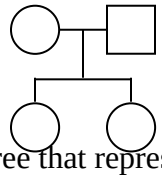
Procedure:

A. First you need to become comfortable in making a pedigree chart. Complete the following examples. You may refer to your Pedigree PowerPoint notes if necessary.

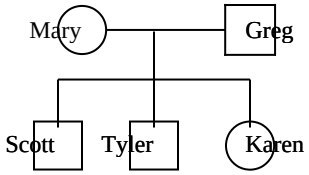
1. How can you tell if a couple is married on a pedigree?
   Write a one-sentence description and draw an example. (1 point for description, 1 point for drawing)
   A circle and a square are connected by a line



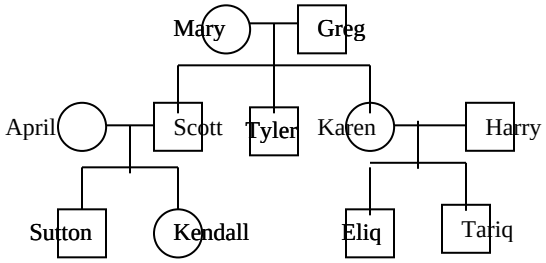
2. How can you tell if the couple who is married had children? Write a one-sentence description and draw an example. (1 point for description, 1 point for drawing)
   The couple has a circle and square connected by a line. The children are connected by a line branching down from the married couple with the children symbols attached to that line.



3. Draw a pedigree that represents Mary married to Greg and with 2 sons (Scott and Tyler) and 1 daughter (Karen). Please label the pedigree with the names of the people. (3 points, each symbol must have labels)

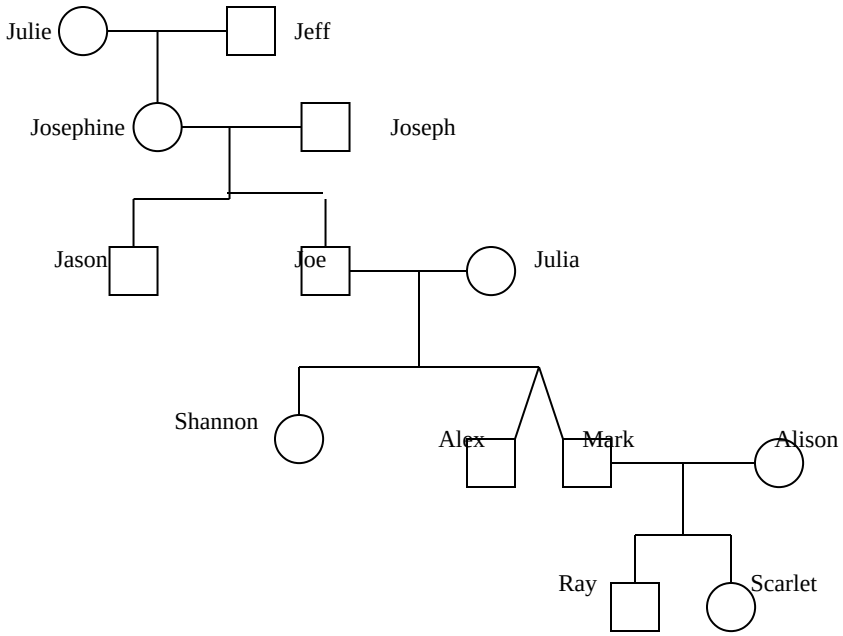


4. Draw a pedigree that represents Mary married to Greg, with 2 sons (Scott and Tyler) and 1 daughter. Their son, Scott, married April and had Sutton (a boy) and Kendall (a girl). Their daughter, Karen, married Harry and had Eliq (a son) and Tariq (a son). Please label the pedigree with the names of the people. (3 points, each symbol must have labels)



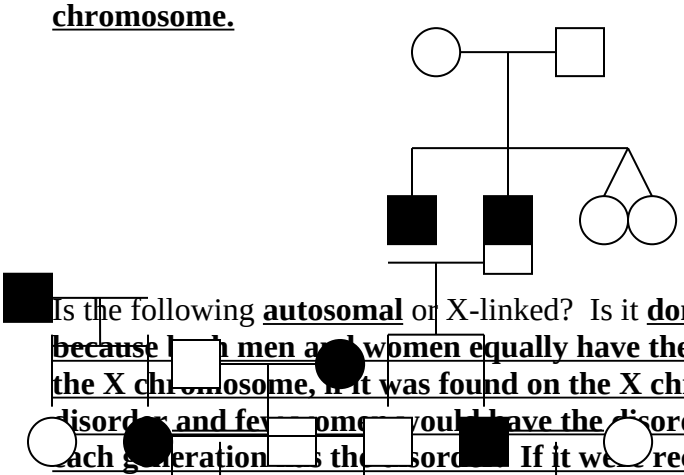
Created by Lauren Almaguer, CDC Science Ambassador, 2004.

5. Draw a pedigree that represents Julie married to Jeff, with one daughter, Josephine. Josephine married Joseph and had Jason and Joe. Joe married Julia and had Shannon and fraternal twin boys, Mark and Alex. Mark married Alison and had Ray and Scarlet. Please label the pedigree with the names of the people. (4 points, each symbol must have labels)



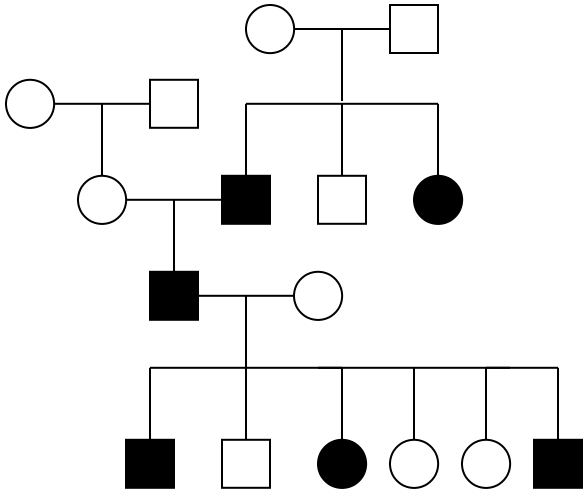
A. Identify the following pedigree charts as autosomal, X-linked, recessive, and dominant. Please explain your answers. (3 points for each correct answer, including correct explanation)

1. Is the following autosomal or **X-linked**? Is it dominant or **recessive**? **The following pedigree is X-linked. One reason is because most of the males in the pedigree have the disorder. This means that the mother is a carrier of the disorder and has given the gene to the sons on the X chromosome. The father cannot give the sons an X chromosome; he only gives them a Y chromosome. This pedigree is recessive due to the fact that neither parent has the disorder. This means the mother can be heterozygous and the father possesses a normal gene on the X chromosome.**



3. Is the following **autosomal** or X-linked? Is it **dominant** or recessive? **This pedigree is autosomal because both men and women equally have the disorder. This means the disorder is not found on the X chromosome, if it was found on the X chromosome, Most of the men would have the disorder and few women would have the disorder. The disorder is dominant, because a parent in each generation has the disorder. If it were recessive the parents would be heterozygous.**
3. Is the following **autosomal** or X-linked? Is it dominant or **recessive**? **This pedigree is autosomal because both men and women equally have the disorder. This means the disorder is not found on the X chromosome. If it were found on the X chromosome, most of the men would have the disorder and few women would have the disorder. This pedigree is recessive due to the fact that**

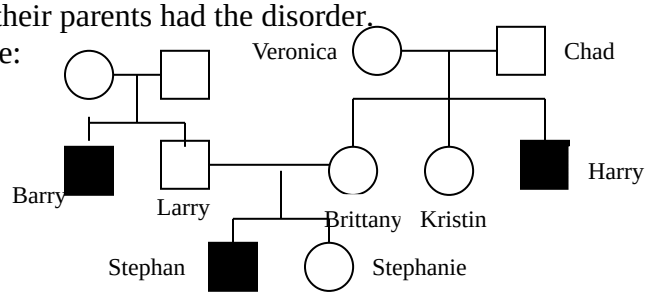
neither parent has the disorder. This means the mother is heterozygous and the father possesses a normal gene on the X chromosome.



B. Now you must make a pedigree chart from the descriptions given. Tell whether it is autosomal or X-linked and whether it is dominant or recessive. Also, state which type of muscular dystrophy the pedigree could be based off of the list your teacher has given you. Label the pedigree with the names of the individuals. (2 points for correct chart including symbols with labels. 2 points for correct description and type of MD.)

a. Chad and Veronica got married and had Brittany, Kristin, and Harry. It was discovered that Harry had muscular dystrophy. Brittany married Larry and had Stephan and Stephanie. Stephan also had muscular dystrophy. Larry's brother Barry also had muscular dystrophy but neither of their parents had the disorder.

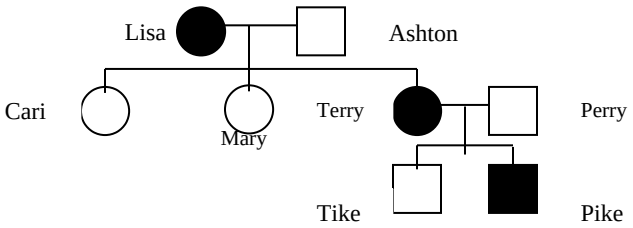
Draw a pedigree:



Is it Autosomal or X-linked? Is it Dominant or recessive? Name the types of muscular dystrophy this pedigree chart could represent. **This is X-linked and recessive. It could be Duchenne or Emery-Dreifuss. These two forms of muscular dystrophy are X-linked and recessive.**

b. Lisa and Ashton got married and had three girls, Cari, Mary, and Terry. It was discovered that Lisa had muscular dystrophy. Terry married Perry and had two boys, Pike and Tike. It was discovered that Terry and Pike had muscular dystrophy.

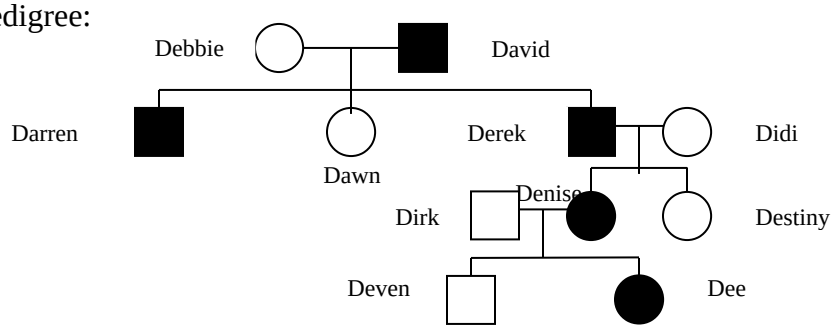
Draw a pedigree



Is it autosomal or X-linked? Is it dominant or recessive? Name the types of muscular dystrophy this pedigree chart could represent**. This is autosomal and dominant. This could be Distal, Emery-Dreifuss, or Limb-Girdle, because these forms of muscular dystrophy are autosomal and dominant**

c. Debbie married David and had three children, Darren, Dawn, and Derek. David, Darren and Derek discovered they have muscular dystrophy. Derek married Didi and had two children, Denise and Destiny. Denise also has muscular dystrophy and married Dirk. They had two children, Dee and Deven. Dee has muscular dystrophy.

Draw a pedigree:



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Is it autosomal or X-linked? Is it dominant or recessive? **This is autosomal and dominant. Name the types of muscular dystrophy this pedigree chart could represent. This pedigree is autosomal because both men and women equally have the disorder. The disorder is dominant, because a parent in each generation has the disorder. If it were recessive, the parents could be heterozygous. This could be distal, Erb's fascioscapulohumeral, Grower's, or limb-girdle, because these forms of muscular dystrophy are autosomal and dominant.**